6-9 Digital Representation and Compression Technology for Videos

Point!

1 Mechanism of Videos

(1) ([1]**Video**) : An electronic medium that creates the illusion of movement by displaying a series of still images in succession. It utilizes the ([2]**afterimage phenomenon**), which occurs due to the characteristics of human vision.

(2) ([3]**Frame**) : Each still image that composes a video.

(3) ([4]**Frame rate**) : The number of frames displayed per second. The unit is ([5]**fps**).
The ([6]**higher**) the frame rate number, the smoother the video will appear, and the ([7]**larger**) the data size will become.

(4) How to Calculate the Amount of Video Data
Amount of video data = ([8]**Amount of image data** [**B**] × **Frame rate**[**fps**] × **Time** [**seconds**])

2 Data Compression

(1) ([9]**Compression**) : A process of reducing the amount of data as much as possible while preserving the content of the data.

(2) ([10]**Decompression**) : A process of restoring compressed data to its original state.

(3) ([11]**Compression ratio**) : The extent to which data has been compressed.

$$\text{Compression ratio [\%]} = \frac{([12]\textbf{Amount of data after compression})}{([13]\textbf{Original amount of data})} \times 100$$

(4) Lossless Compression and Lossy Compression

[1] ([14]**Lossless compression**) : A compression method that allows for the complete restoration of the original data from the compressed data. Used for compressing information such as text or program data.

[2] ([15]**Lossy compression**) : A compression method in which the original data cannot be restored from the compressed data. Lossy compression is used for compressing audio, images, and video files in a way that humans do not perceive a significant difference, even if those files cannot be perfectly restored to their original form.

3 Main Types of Lossless Compression

(1) ([16]**Run-length encoding**) : A compression method that replaces sequences of the same consecutive symbols with a numerical value indicating the length of the sequence. It is effective when there are many consecutive identical symbols.

<Example> Before compression: "AAAAABBAAAABBBBBBBBAAAAAA" (25 characters)

↓

AAAAA (5 of "A") BB (2 of "B") AAAA (4 of "A") BBBBBBBB (8 of "B") AAAAAA (6 of "A")

After compression: ([17]**A5B2A4B8A6**) (10 characters)

The compression ratio is ([18] $\frac{10}{25} \times 100$) = ([19]**40 [%]**).

(2) ([20]**Huffman coding**) : A compression method that assigns shorter bit sequences to characters with higher frequencies of occurrence, and longer bit sequences to those with lower frequencies.

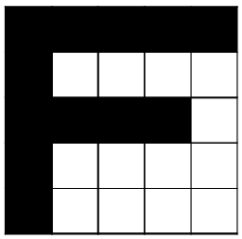
Warm Up

Answer the following questions.

(1) [1] What is the data size, in megabytes, of a 10-second uncompressed video at 30 fps, where each frame is a 24-bit full-color still image with a resolution of 500 × 200 pixels? For this question, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

[2] Video [1] was converted into a compressed video file using a certain compression method, resulting in a file size of 30 MB. What is the compression ratio in this conversion? Round your answer to the nearest whole number and provide the answer as an integer.

(2) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBBBWWWWBBBBWBWWWWBWWWW".
Next, by representing consecutive characters "B" or "W" of length n (where n is an integer of 2 or more) as "Bn" or "Wn", the sequence can be compressed to "B6W4B4WBW4BW4". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?



"B4WBW3B5WBW4BW4"

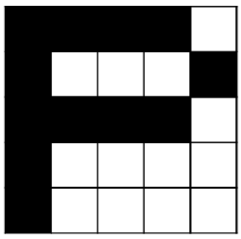
Explanation

(1) [1] The amount of data per frame is,
500 × 200 × 24 [bits] = 2,400,000 [bits]
Since 1 B = 8 bits, 2,400,000 [bits] ÷ 8 = 300,000 [B]
The amount of data of a 10-second video at 30 fps is as follows:
300,000 [B] × 30 [fps] × 10 [seconds] = 90,000,000 [B]
1 KB = 1,000 B, 1 MB = 1,000 KB, therefore,
90,000,000 [B] ÷ 1,000 ÷ 1,000 = 90 [MB] 90 MB

[2] Compression ratio [%] = $\frac{\text{Amount of data after compression}}{\text{Original amount of data}} \times 100$,

resulting in $\frac{30}{90} \times 100 = 33.3\cdots$ [%] 33%

(2) Expanding the pre-compressed data "B4WBW3B5WBW4BW4" yields "BBBBWBWWWBBBBBWBWWWWBWWWW".
If you fill in a 5 × 5 grid one pixel at a time from the top-left, row by row to the right, coloring black for "B" and white for "W", you will get the image shown on the right. Therefore, P



Try

1 Answer the following questions. For these questions, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

(1) Complete the following sentences by filling in the blanks [1] and [5] with the appropriate terms.

> A video is a continuous display of still images. By displaying a series of gradually changing images, the phenomenon of ([1]) creates the illusion of movement for humans. The individual still images that make up a video are called ([2]), and the number of ([2]) displayed on the screen per second is referred to as the ([3]). The ([4]) the ([3]), the smoother the motion in the video appears. However, the amount of data of videos can become ([5]).

(2) What is the data size, in megabytes, of a 1-minute video at 30 fps, where each frame is a 24-bit full-color still image with a resolution of 640 × 360 pixels? Assume that the video is uncompressed. Round the amount of data to the nearest whole number and provide the answer as an integer.

2 Answer the following questions.

(1) For the following statements **A** to **D** about data compression, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Lossless compression is a method that improves compression efficiency by allowing slight changes to the data, such as removing components that are less noticeable to humans.

B Decompression is a process of restoring compressed data to its original state.

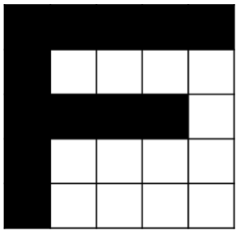
C Data compressed using lossy compression does not result in exactly the same content as the original uncompressed data.

D Data compressed using lossless compression will be exactly the same as the original data after being decompressed.

(2) When 21 MB of compressed data was decompressed, it expanded to 50 MB. What is the compression ratio of this file?

3 Answer the following questions.

(1) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBBBWWWWBBBBWBWWWWBWWWW".
Next, by representing consecutive characters "B" or "W" of length n (where n is an integer of 2 or more) as "Bn" or "Wn", the sequence can be compressed to "B6W4B4WBW4BW4". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?
"BW3B2W3B7W3B2W3B"



(2) What is the term for the compression method that represents repeated data by listing the data and the number of consecutive occurrences, as shown in (1)? Choose one from the options **A** to **D**, and answer using the letter.

A Huffman coding **B** Codec

C Run-length encoding **D** LZ encoding

Exercise

1 Answer the following questions. For these questions, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB.

(1) For the following statements **A** to **D** about video mechanisms, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Videos create the illusion of movement by displaying a series of still images in rapid succession, leveraging the afterimage phenomenon. In reality, the images themselves are not moving.

B Each individual still image that composes a video is called a frame.

C The number of frames displayed per second is called the frame rate. The lower the frame rate, the smoother the video appears.

D When recording a video, if the resolution and frame rate settings are the same, the amount of data of a video recorded for 1 second and a video recorded for 60 seconds does not vary.

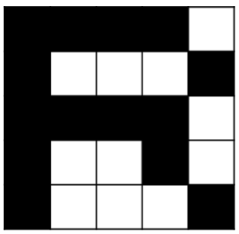
(2) What is the data size, in megabytes, of a 1-second video at 60 fps, where each frame is a 24-bit full-color still image with a resolution of 1,920 × 1,080 pixels? Assume that the video is uncompressed, and round your answer to one decimal place.

2 For the terms that fit in the blanks [1] to [3] in the following sentences, if the compression is lossless, answer with **A**; if the compression is lossy, answer with **B**.

> There are two main methods for compressing data. A compression method that allows the original data to be completely restored from the compressed data when decompressed is called ([1]). A compression method that does not allow for the complete restoration of the original data is called ([2]). For data such as images and videos, where the compressed data does not need to be perfectly restored and humans do not perceive a significant difference, ([3]) is used.

3 Answer the following questions.

(1) Consider how to represent an image using pixels. In the case of the diagram on the right, the 25 pixels in the 5 × 5 grid are read one by one from the top-left, row by row, moving to the right. If a pixel is black, write "B"; if it is white, write "W". This results in the sequence:
"BBBBWBWWWBBBBBWBWWBWBWWWB".
When "B" or "W" appears consecutively n times (where n is an integer greater than or equal to 2), they can be written as "Bn" or "Wn". Using this method, the sequence can be compressed as "B4WBW3B5WBW2BWBW3B". Using the same method with another 5 × 5 grid, what is the alphabet letter represented by the following compressed data?
"BW3BWBWBW3BW4BW4BW2"



(2) What is the term for the compression method that represents repeated data by listing the data and the number of consecutive occurrences, as shown in (1)?